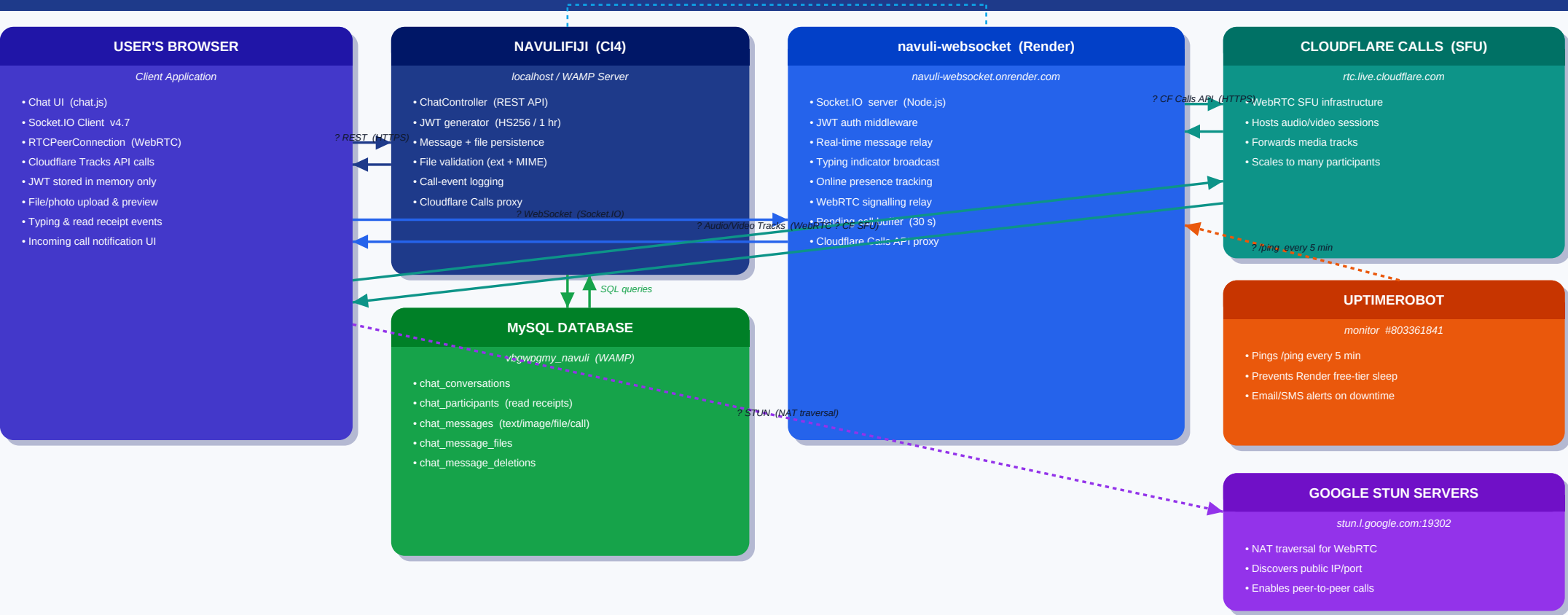


System Architecture

Chat • Voice Call • Video Call

This document describes the full architecture of the real-time communication features integrated into Navuli Fiji — covering all external services, data flows, security mechanisms and the role of each component.





FLOW LEGEND

- REST API (HTTP/S)**

Browser ? CI4 ? MySQL: send & fetch messages, login, upload files, get JWT token for Socket.IO
- CF Calls API (Proxy)**

Render ? Cloudflare: create sessions, publish/pull audio-video tracks; CF_API_TOKEN stays server-side
- STUN (NAT Traversal)**

Browser ? Google STUN: discovers public IP & port so browsers can reach each other for peer-to-peer calls
- WebSocket (Socket.IO)**

Browser ? Render: real-time delivery, typing, online status, call signalling, read receipts
- Audio/Video (WebRTC)**

Browser ? Cloudflare SFU: actual media streams; Render only relays the session/track metadata (signals)
- Uptime Monitoring**

UptimeRobot pings /ping every 5 min ? keeps Render free-tier awake ? sends email/SMS alert on downtime

Service	URL / Location	Technology	Role in the System
NavuliFiji (CI4)	localhost/navulifiji/public	PHP 8 / CodeIgniter 4	Main web app: user auth, REST API, JWT token generation, message & file storage, call logging, CF Calls proxy
navuli-websocket	navuli-websocket.onrender.com	Node.js 22 + Socket.IO 4	Real-time relay: messages, typing indicators, online presence, WebRTC call signalling, CF Calls API proxy
MySQL Database	localhost:3306 (WAMP)	MySQL 8 / InnoDB / utf8mb4	Persistent storage: conversations, participants, messages (text/image/file/call), file metadata, per-user soft-deletes
Cloudflare Calls	rtc.live.cloudflare.com	WebRTC SFU (Cloudflare)	Audio/video infrastructure: hosts call sessions, receives media tracks from caller, forwards them to callee at scale
UptimeRobot	dashboard.uptimerobot.com	Monitoring SaaS	Pings /ping every 5 min to prevent Render free-tier sleep; sends email/SMS alerts if server goes down
Google STUN	stun.l.google.com:19302	STUN Protocol (RFC 5389)	NAT traversal: helps browsers discover their public address and port so WebRTC peer-to-peer connections can be established

Security Architecture

JWT Authentication

Socket.IO connections authenticated with HS256 JWT (1-hour expiry). CI4 generates the token from CHAT_JWT_SECRET; the Render server validates it using the same secret. Mismatch = connection rejected.

File Upload Validation

Files validated by extension AND MIME type. Max 10 MB enforced. Filenames replaced with 32-byte cryptographically random hex strings. Participant membership verified before any upload is accepted.

Uptime & Availability

UptimeRobot monitors /ping every 5 minutes, preventing Render free-tier spin-down (15-min idle timeout). Email and SMS alerts fire immediately on downtime so action can be taken quickly.

Cloudflare API Token

CF_API_TOKEN never reaches the browser. All Cloudflare Calls API calls are proxied through the Render server (requireJwt middleware protects every endpoint), keeping the token strictly server-side.

Message Access Control

Every REST endpoint calls isParticipant() before reading or writing. "Remove for me" soft-deletes are per-user. "Remove for everyone" (hard-delete) is restricted exclusively to the original sender.

HTTPS / WSS Everywhere

Render serves over TLS (HTTPS + WSS). Cloudflare Calls uses HTTPS. WAMP runs locally only (no public exposure). No sensitive tokens or credentials travel over plain HTTP in any flow.